SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

*I **Prudhvi Raj.B(192324274)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prudhvi raj.b

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

(5 Marks)

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is

the probability that the car came from Section C? (5 Marks)

3. In a real-world medical image classification task for detecting breast cancer, a deeplearning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier,

explaining its effectiveness in medical diagnosis. (5 Marks)

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit. ii) Suggest two effective techniques to reduce overfitting in this model. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63% (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70% (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68% i) Identify the epoch at which the model starts overfitting.
ii) Explain why validation accuracy decreases despite training accuracy increasing. iii) Suggest two methods to improve validation performance.

6. A machine learning model shows the following loss values during training: (5 Marks)

(a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
(b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3 (c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
(d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
(e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
(f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8 (h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
i) At which epoch does the model achieve optimal generalization?
ii)Identify signs of overfitting from the data. iii) Recommend two regularization techniques and explain their impact.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

a) Front identified as Front: 119
b) Front identifies as Back: 06
c) Back identified as Back: 120
d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a) Diseased identified as Diseased: 138
b) Diseased identified as Healthy: 12
c) Healthy identified as Healthy: 135
d) Healthy identified as Diseased: 15

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks)

- Defective identified as Defective: 2 30
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.

(5 Marks)

12. A warehouse has two sections. Section A contains 15 defective items and 35 nondefective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.

(5 Marks)

13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150.The dataset contains 1000 samples per class, and 20% of the data is used for testing. (5 Marks)

Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data.

Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

15. A deep learning model is trained for a classification task, and the following loss values are observed: (5 Marks)

i. Epoch 1: Training Loss = 2.8, Validation Loss = 2.6 ii.
Epoch 2: Training Loss = 2.4, Validation Loss = 2.2 iii.
Epoch 3: Training Loss = 2.1, Validation Loss = 1.9 iv.
Epoch 4: Training Loss = 1.9, Validation Loss = 1.8 v.
Epoch 5: Training Loss = 1.7, Validation Loss = 1.9 vi.
Epoch 6: Training Loss = 1.5, Validation Loss = 2.1 vii.
Epoch 7: Training Loss = 1.3, Validation Loss = 2.4
viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7

Answer the following:

a. At which epoch does the model achieve the best generalization performance?
b. At which point does overfitting begin? Justify your answer.
c. Suggest two techniques to reduce overfitting and briefly explain how they help.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Question 1 — Bayes' Theorem: Colored Balls in Boxes

Box A: 6 red, 4 blue, 2 green (12 total). Box B: 5 red, 7 blue, 3 green (15 total). Box C: 8 red, 2 blue, 5 green (15 total). Box B is twice as likely to be chosen as A or C. A blue ball is drawn. Find P(Box C | Blue).

Solution

Let $P(A) = P(C) = x$ and $P(B) = 2x$. Since they must sum to 1: $x + 2x + x = 1 \rightarrow 4x = 1 \rightarrow x = 1/4$.

So $P(A) = 1/4$, $P(B) = 1/2$, $P(C) = 1/4$.

$P(\text{Blue} \mid A) = 4/12 = 1/3$ $P(\text{Blue} \mid B) = 7/15$ $P(\text{Blue} \mid C) = 2/15$

Total probability of drawing blue:

$P(\text{Blue}) = (1/4)(1/3) + (1/2)(7/15) + (1/4)(2/15) = 5/60 + 14/60 + 2/60 = 21/60 = 7/20 = 0.35$

By Bayes' Theorem:

$P(C \mid \text{Blue}) = [P(C) \times P(\text{Blue}|C)] / P(\text{Blue}) = [(1/4)(2/15)] / (7/20) = (1/30) / (7/20) = 2/21$

Answer: P(Box C | Blue) = 2/21 ≈ 0.0952 (about 9.52%)

Question 2 — Bayes' Theorem: Cars by Section

Section A: 5 red, 3 blue, 2 black (10 total). Section B: 4 red, 6 blue, 5 black (15 total). Section C: 7 red, 2 blue, 6 black (15 total). Section B is twice as likely to be chosen. A blue car is drawn. Find P(Section C | Blue).

Solution

As before, $P(A) = P(C) = 1/4$, $P(B) = 1/2$.

$P(\text{Blue} \mid A) = 3/10$ $P(\text{Blue} \mid B) = 6/15 = 2/5$ $P(\text{Blue} \mid C) = 2/15$

$P(\text{Blue}) = (1/4)(3/10) + (1/2)(2/5) + (1/4)(2/15) = 9/120 + 24/120 + 4/120 = 37/120$

$P(C \mid \text{Blue}) = [(1/4)(2/15)] / (37/120) = (1/30) / (37/120) = 4/37$

Answer: P(Section C | Blue) = 4/37 ≈ 0.1081 (about 10.81%)

Question 3 — Confusion Matrix: Breast Cancer Classification (KNN)

Dataset: 1800 samples per class; 20% reserved for testing → test set = 20% × 3600 = 720 samples (360 per class). Given: TP = 210, TN = 190, FP = 30.

Step 1: Find False Negatives (FN)

FN = Total Test Samples − (TP + TN + FP) = 720 − (210 + 190 + 30) = 720 − 430 = 290

Metric / Item	Value
True Positive (TP)	210
True Negative (TN)	190
False Positive (FP)	30
False Negative (FN)	290
Total Test Samples	720

Step 2: Compute Metrics

Accuracy = (TP + TN) / Total = (210 + 190) / 720 = 400/720 = 0.5556 → 55.56%

Precision = TP / (TP + FP) = 210 / 240 = 0.875 → 87.5%

Sensitivity (Recall) = TP / (TP + FN) = 210 / 500 = 0.42 → 42%

Specificity = TN / (TN + FP) = 190 / 220 = 0.8636 → 86.36%

Interpretation

The model shows very high Precision (87.5%) and Specificity (86.36%), meaning it rarely mislabels benign cases as malignant. However, Sensitivity is low (42%), meaning it misses more than half of actual malignant cases (high FN). In medical diagnosis, low sensitivity is dangerous because missed cancer cases (false negatives) can delay critical treatment; the model would need to be tuned (e.g., threshold adjustment) to raise recall even at some cost to precision.

Question 4 — Overfitting Analysis from Loss Curves

Epoch	1	2	3	4	5	6	7	8
Training Loss	2.8	2.5	2.2	2	1.8	1.6	1.5	1.3
Validation Loss	2.4	2.2	2	2.1	2.3	2.6	2.9	3.2

i) Epoch where overfitting begins

Validation loss keeps falling with training loss through Epoch 3 (its lowest point, 2.0). From Epoch 4 onward, training loss keeps decreasing but validation loss starts rising (2.1, 2.3, 2.6...). This divergence marks the onset of overfitting.

Answer: Overfitting begins at Epoch 4.

ii) Two techniques to reduce overfitting

- Early Stopping — halt training once validation loss stops improving (around Epoch 3), preventing the model from over-learning training noise.
- Dropout / L2 Regularization (weight decay) — randomly deactivating neurons or penalizing large weights reduces model complexity and forces more generalizable feature learning.

iii) Effect of overfitting on generalization

Overfitting occurs when a model learns the training data's noise and specific patterns instead of the underlying general trend. This causes very low training error but rising error on unseen (validation/test) data — the model's ability to generalize deteriorates, resulting in poor real-world performance despite strong training performance.

Question 5 — Overfitting Analysis from Accuracy Curves

Epoch	1	2	3	4	5	6	7	8
Training Accuracy (%)	60	65	70	75	80	85	88	92
Validation Accuracy (%)	58	63	68	72	73	72	70	68

i) Epoch where overfitting starts

Validation accuracy peaks at Epoch 5 (73%) and then declines steadily (72%, 70%, 68%) even though training accuracy keeps rising.

Answer: Overfitting starts at Epoch 6.

ii) Why validation accuracy decreases while training accuracy increases

As training continues, the model increasingly memorizes fine-grained patterns and noise specific to the training set rather than learning generalizable features. This increases the gap between training and validation performance (high variance) — the model fits training data more closely but loses its ability to correctly classify unseen validation samples.

iii) Two methods to improve validation performance

- Early Stopping — stop training near Epoch 5, where validation accuracy is highest, before performance degrades.
- Data Augmentation / Regularization (Dropout, L2) — increases effective training diversity or constrains model complexity, helping the model generalize better to validation/unseen data.

Question 6 — Optimal Generalization and Regularization

Epoch	1	2	3	4	5	6	7	8
Training Loss	3	2.6	2.3	2.1	1.9	1.7	1.6	1.4
Validation Loss	2.7	2.3	2.1	2	2.2	2.5	2.8	3.1

i) Epoch of optimal generalization

Validation loss is lowest at Epoch 4 (2.0), the point where the model best balances fitting the training data with generalizing to unseen data.

Answer: Optimal generalization occurs at Epoch 4.

ii) Signs of overfitting

From Epoch 5 onward, training loss continues to decrease (1.9 → 1.4) while validation loss steadily increases (2.2 → 3.1). This growing gap between training and validation loss is the classic signature of overfitting.

iii) Two regularization techniques and their impact

- L2 Regularization (Weight Decay) — adds a penalty proportional to the square of the weights to the loss function, discouraging overly large weights and producing a simpler, smoother decision boundary that generalizes better.
- Dropout — randomly drops a fraction of neurons during each training pass, preventing neurons from co-adapting and forcing the network to learn redundant, more robust feature representations.

Question 7 — Confusion Matrix: Tire Segregation (250 tires)

Metric / Item	Value
Front identified as Front (TP)	119
Front identified as Back (FN)	6
Back identified as Back (TN)	120
Back identified as Front (FP)	5
Total	250

Calculations

Accuracy = (TP + TN) / Total = (119 + 120) / 250 = 239/250 = 0.956 → 95.6%

Precision = TP / (TP + FP) = 119 / 124 = 0.9597 → 95.97%

Sensitivity (Recall) = TP / (TP + FN) = 119 / 125 = 0.952 → 95.2%

Specificity = TN / (TN + FP) = 120 / 125 = 0.96 → 96%

F1-Score = 2 × (Precision × Recall) / (Precision + Recall) = 2 × (0.9597 × 0.952) / (0.9597 + 0.952) = 1.8272 / 1.9117 ≈ 0.9558 → 95.58%

Question 8 — Confusion Matrix: Disease Diagnosis (300 cases)

Metric / Item	Value
Diseased identified as Diseased (TP)	138
Diseased identified as Healthy (FN)	12
Healthy identified as Healthy (TN)	135
Healthy identified as Diseased (FP)	15
Total	300

Calculations

Accuracy = (138 + 135) / 300 = 273/300 = 0.91 → 91%

Precision = 138 / (138 + 15) = 138/153 = 0.9020 → 90.20%

Sensitivity (Recall) = 138 / 150 = 0.92 → 92%

Specificity = 135 / 150 = 0.90 → 90%

F1-Score = 2 × (0.9020 × 0.92) / (0.9020 + 0.92) = 1.6596 / 1.8220 ≈ 0.9109 → 91.09%

Question 9 — Confusion Matrix: Spam Email Filtering (400 emails)

Metric / Item	Value
Spam identified as Spam (TP)	180
Spam identified as Not Spam (FN)	20
Not Spam identified as Not Spam (TN)	170
Not Spam identified as Spam (FP)	30
Total	400

Calculations

Accuracy = (180 + 170) / 400 = 350/400 = 0.875 → 87.5%

Precision = 180 / (180 + 30) = 180/210 = 0.8571 → 85.71%

Sensitivity (Recall) = 180 / 200 = 0.90 → 90%

Specificity = 170 / 200 = 0.85 → 85%

F1-Score = 2 × (0.8571 × 0.90) / (0.8571 + 0.90) = 1.5429 / 1.7571 ≈ 0.8780 → 87.80%

Question 10 — Confusion Matrix: Quality Inspection (500 products)

Metric / Item	Value
Defective identified as Defective (TP)	230
Defective identified as Non-Defective (FN)	20
Non-Defective identified as Non-Defective (TN)	225
Non-Defective identified as Defective (FP)	25
Total	500

Calculations

Accuracy = (230 + 225) / 500 = 455/500 = 0.91 → 91%

Precision = 230 / (230 + 25) = 230/255 = 0.9020 → 90.20%

Sensitivity (Recall) = 230 / 250 = 0.92 → 92%

Specificity = 225 / 250 = 0.90 → 90%

F1-Score = 2 × (0.9020 × 0.92) / (0.9020 + 0.92) ≈ 0.9109 → 91.09%

Question 11 — Bayes' Theorem: Oil Bottles Across Shops

Shop X: 20 pure, 30 adulterated (50 total). Shop Y: 40 pure, 20 adulterated (60 total). Shop Y is twice as likely to be chosen as Shop X. A selected bottle is found adulterated. Find the probability it came from each shop.

Solution

Let P(X) = x, P(Y) = 2x → 3x = 1 → x = 1/3. So P(X) = 1/3, P(Y) = 2/3.

P(Adulterated | X) = 30/50 = 3/5   P(Adulterated | Y) = 20/60 = 1/3

P(Adulterated) = (1/3)(3/5) + (2/3)(1/3) = 1/5 + 2/9 = 9/45 + 10/45 = 19/45

By Bayes' Theorem:

P(X | Adulterated) = [(1/3)(3/5)] / (19/45) = (1/5) / (19/45) = 9/19 ≈ 0.4737

P(Y | Adulterated) = [(2/3)(1/3)] / (19/45) = (2/9) / (19/45) = 10/19 ≈ 0.5263

Answer: P(Shop X | Adulterated) = 9/19 ≈ 47.37%, P(Shop Y | Adulterated) = 10/19 ≈ 52.63%

Question 12 — Bayes' Theorem: Defective Items in Warehouse Sections

Section A: 15 defective, 35 non-defective (50 total). Section B: 25 defective, 25 non-defective (50 total). Section A is three times as likely to be chosen as Section B. A selected item is defective. Find the probability it came from each section.

Solution

Let P(B) = x, P(A) = 3x → 4x = 1 → x = 1/4. So P(A) = 3/4, P(B) = 1/4.

P(Defective | A) = 15/50 = 0.3   P(Defective | B) = 25/50 = 0.5

P(Defective) = (3/4)(0.3) + (1/4)(0.5) = 0.225 + 0.125 = 0.35

By Bayes' Theorem:

P(A | Defective) = (0.75 × 0.3) / 0.35 = 0.225 / 0.35 = 9/14 ≈ 0.6429

P(B | Defective) = (0.25 × 0.5) / 0.35 = 0.125 / 0.35 = 5/14 ≈ 0.3571

Answer: P(Section A | Defective) = 9/14 ≈ 64.29%, P(Section B | Defective) = 5/14 ≈ 35.71%

Question 13 — Confusion Matrix: KNN Binary Classification

1000 samples per class; 20% used for testing → 200 test samples per class (400 total). Given TP = 120, TN = 150.

Solution

Actual positives in test set = 200, so FN = 200 − TP = 200 − 120 = 80.

Actual negatives in test set = 200, so FP = 200 − TN = 200 − 150 = 50.

Metric / Item	Value
TP	120
TN	150
FP	50
FN	80
Total	400

Accuracy = (120 + 150) / 400 = 270/400 = 0.675 → 67.5%

Precision = 120 / (120 + 50) = 120/170 = 0.7059 → 70.59%

Sensitivity (Recall) = 120 / (120 + 80) = 120/200 = 0.60 → 60%

Specificity = 150 / (150 + 50) = 150/200 = 0.75 → 75%

Question 14 — Confusion Matrix: Medical Image Classification (KNN)

1200 samples per class; 25% used for testing → 300 test samples per class (600 total). Given TP = 180, TN = 160.

Solution

Actual positives = 300, so FN = 300 − TP = 300 − 180 = 120.

Actual negatives = 300, so FP = 300 − TN = 300 − 160 = 140.

Metric / Item	Value
TP	180
TN	160
FP	140
FN	120
Total	600

Accuracy = (180 + 160) / 600 = 340/600 = 0.5667 → 56.67%

Precision = 180 / (180 + 140) = 180/320 = 0.5625 → 56.25%

Sensitivity (Recall) = 180 / (180 + 120) = 180/300 = 0.60 → 60%

Specificity = 160 / (160 + 140) = 160/300 = 0.5333 → 53.33%

Question 15 — Best Generalization Point and Overfitting Onset

Epoch	1	2	3	4	5	6	7	8
Training Loss	2.8	2.4	2.1	1.9	1.7	1.5	1.3	1.2
Validation Loss	2.6	2.2	1.9	1.8	1.9	2.1	2.4	2.7

a) Epoch of best generalization

Validation loss is at its minimum (1.8) at Epoch 4, indicating the best balance between bias and variance.

Answer: Best generalization performance occurs at Epoch 4.

b) Where does overfitting begin?

From Epoch 5 onward, training loss keeps falling (1.7 → 1.2) while validation loss rises (1.9 → 2.7). This divergence — training error still dropping but validation error climbing — is the hallmark of overfitting.

Answer: Overfitting begins at Epoch 5.

c) Two techniques to reduce overfitting

- Early Stopping — stop training at Epoch 4, where validation loss is lowest, to prevent the model from continuing to fit noise in later epochs.
- Dropout Regularization — randomly deactivates a fraction of neurons during training, reducing reliance on specific neurons/co-adaptation and improving the model's ability to generalize to unseen data.